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Next item →

1. Let U be the universal set and A be a subset of U . Which one of the following is TRUE?

☐ $\overline{A} \cup A = U$

☐ $\overline{A} = U$

☐ $\overline{A} - A = \emptyset$

☐ $\overline{A} - A = U$

☒ Nice work

$\overline{A} = U - A$, hence $A \cup \overline{A} = U$

1 / 1 point

2. If the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and a set $A = \{1, 3, 5, 7, 9\}$, the $\overline{A}$ is

☐ $\{2, 4, 6, 8, 10\}$

☐ $\{4, 6, 8, 10\}$

☐ $\{2\}$

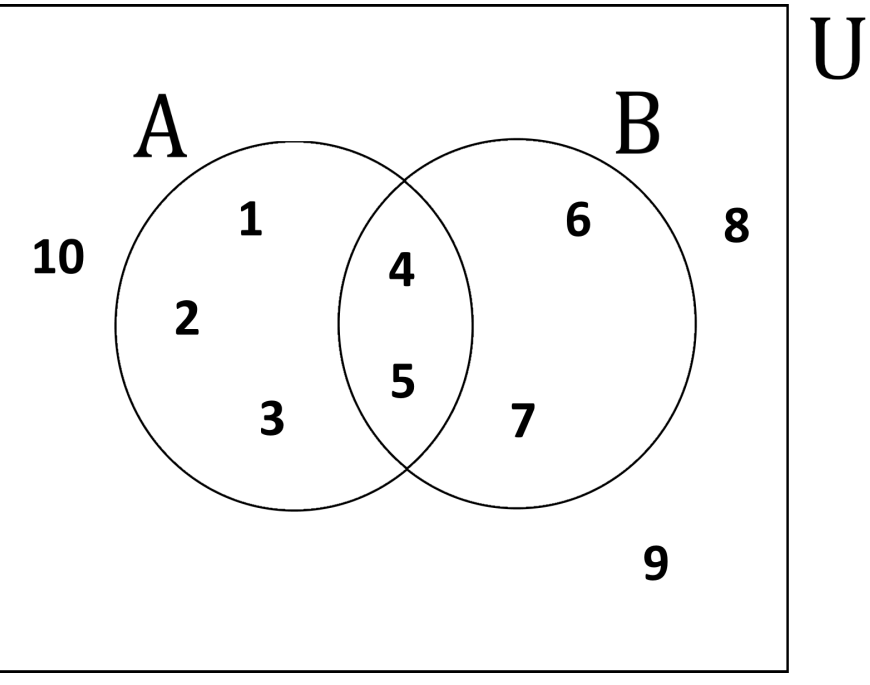
☐ None of these answers

☒ Nice work

$\overline{A} = U - A = \{2, 4, 6, 8, 10\}$

1 / 1 point

3. Let A and B be two subsets of the universal set U , represented by the following Venn diagram:



(Image description: A Venn diagram of two circles, A and B . A contains the numbers 1, 2, 3, 4 and 5. B contains the numbers 4, 5, 6 and 7. The intersection of A and B contains the numbers 4 and 5. A box, labelled U , surrounds both circles. Numbers 8, 9 and 10 are outside circles A and B but within box U .)

Which one of the following sets represents $(\overline{A} \oplus \overline{B})$?

☐ $\{8, 9, 10\}$

☐ $\{1, 2, 3, 6, 7\}$

☐ $\{4, 5\}$

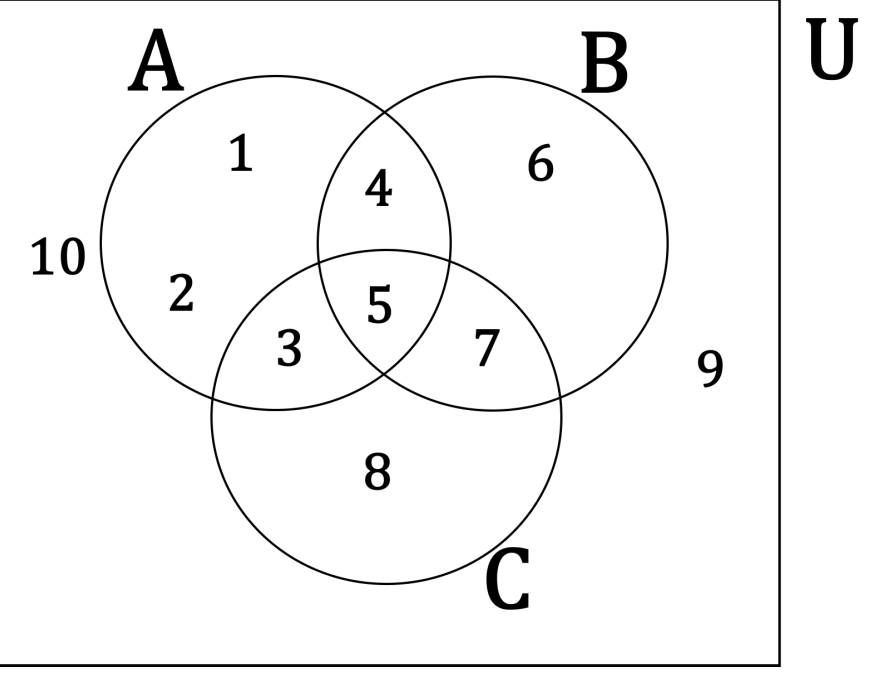
☒ $\{4, 5, 8, 9, 10\}$

☐ Nice work

$A \oplus B = \{1, 2, 3, 6, 7\}$, hence $(\overline{A \oplus B}) = \{4, 5, 8, 9, 10\}$

1 / 1 point

4. Let A , B and C be three subsets of the universal set U , represented by the following Venn diagram:



(Image description: A Venn diagram with 3 circles A , B and C . A contains the numbers 1, 2, 3, 4 and 5. B contains the numbers 4, 5, 6 and 7. C contains the numbers 3, 5, 7 and 8. The intersection of A and B contains the numbers 4 and 5. The intersection of B and C contains the numbers 7 and 8. The intersection of C and A contains the numbers 3 and 5. The intersection of A , B and C contains the number 5. All three circles are set in a rectangle, labelled U . Numbers 9 and 10 are outside the circles and within the box.)

Which one of the following sets represents $(\overline{A} \oplus (\overline{B \cap C}))$?

☐ $\{1, 2, 6, 7, 8\}$

☐ $\{8, 10\}$

☒ $\{3, 4, 5, 9, 10\}$

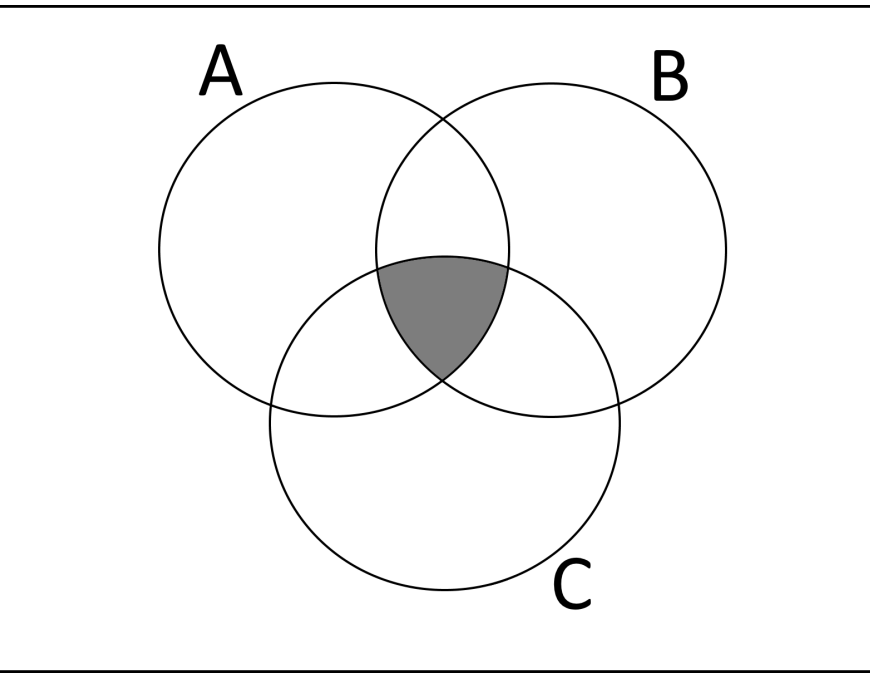
☐ $\emptyset$

☐ Nice work

$A = \{1, 2, 3, 4, 5\}$ and $\overline{B \cap C} = \{8, 4, 5, 6, 7, 8\}$, hence $A \oplus (\overline{B \cap C}) = \{1, 2, 6, 7, 8\}$.
Therefore $(\overline{A \oplus (\overline{B \cap C})}) = \{3, 4, 5, 9, 10\}$

1 / 1 point

5. Let A , B and C be three subsets of the universal set U . Which one of the following represents the area coloured in gray in the Venn diagram below?



(Image description: A Venn diagram showing three circles: A , B and C . The shaded area is the intersection between A , B and C .)

☐ $(A \cup B) - C$

☒ $(A \cap B) \cap C$

☐ $(A \cup B) \oplus C$

☐ $(A \cup B) \cap C$

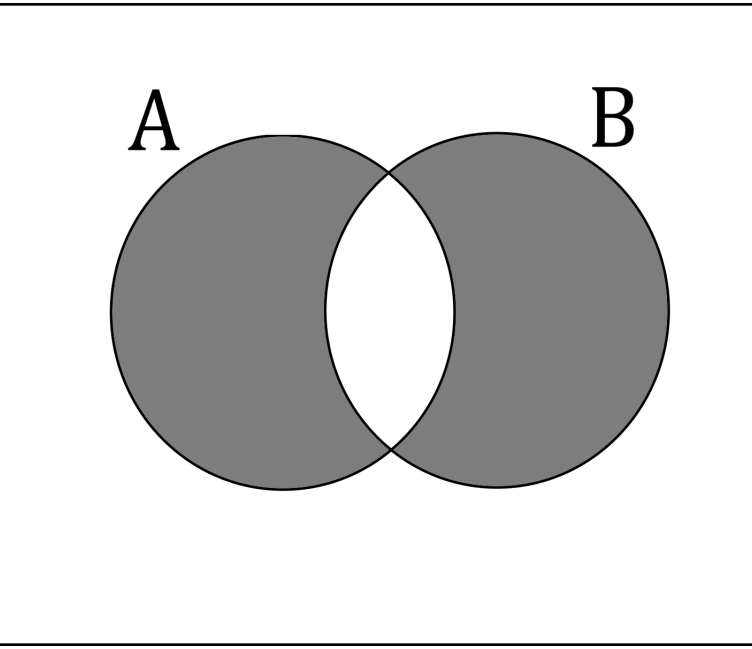
☐ Nice work

The shaded area represents the intersection of the sets A , B and C

1 / 1 point

6. Let A and B be two subsets of the universal set U . Which one of the following Venn diagrams has a shaded (grey) area that represents $A \oplus B$?

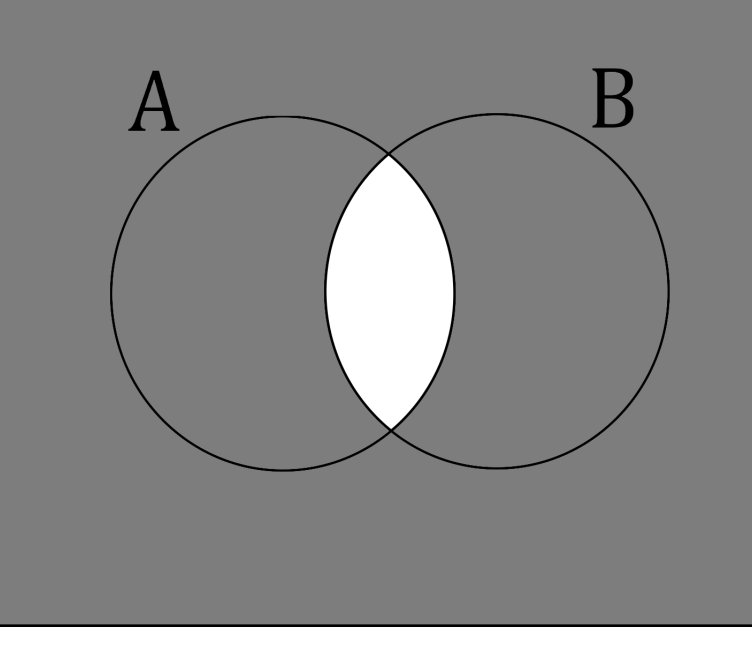
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(Image description: A Venn diagram showing two overlapping circles, A and B , set within a rectangle, U . Circles A and B are shaded. The intersection between A and B , and the rectangle U , are white.)

1 / 1 point

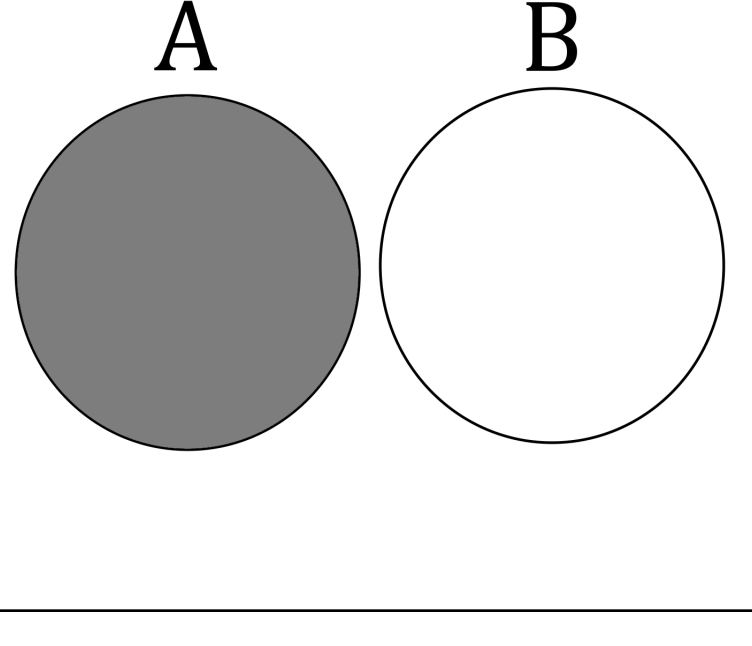
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(Image description: A Venn diagram showing two overlapping circles, A and B , set within a rectangle, U . The intersection between A and B is white. The rest of the diagram is shaded.)

1 / 1 point

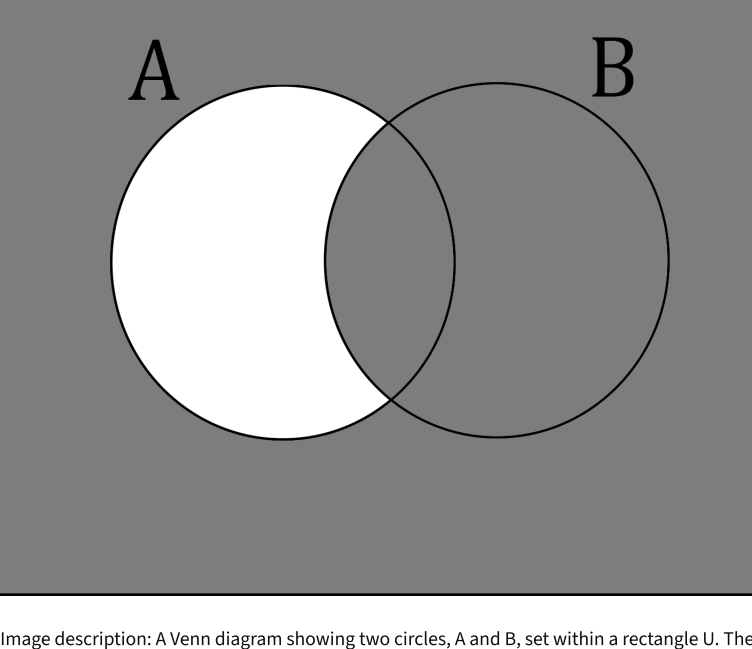
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(Image description: A Venn diagram showing two non-overlapping circles, A and B , set within a rectangle, U . The only unshaded area is circle A . Rectangle U , circle B and the intersection between A and B are shaded.)

1 / 1 point

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(Image description: A Venn diagram showing two circles, A and B , set within a rectangle, U . The only unshaded area is circle A . Rectangle U , circle B and the intersection between A and B are shaded.)

1 / 1 point

https://www.coursera.org/learn/mathematical-foundations-for-computing/assignment-submission/y56Wtbe-the-representation-of-a-set-using-venn-diagrams/view-feedback

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